

Project Title	Rising Waters: Machine Learning Based Flood Prediction System		
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9. Proposed Solution

Bringing the previous design-thinking steps together, the proposed solution below states, in a single canvas, what is being built, why it is different from a plain prediction script, and how it could grow beyond its current scope.

Parameter	Description
Problem Statement	Lack of an accessible, instant flood-risk assessment tool based on historical meteorological data.
Idea / Solution	A web application (Rising Waters) where users input or auto-fill weather parameters and receive an MLgenerated flood-risk classification with explanatory insights.
Novelty / Uniqueness	Combines automatic best-model selection across four algorithms with rule-based, human-readable reasoning layered on top of the raw prediction, rather than exposing a bare probability.
Social Impact	Supports early awareness and preparedness for residents and local monitoring bodies in flood-prone regions.
Business Model	Currently an educational/demonstrative solo project; not commercialized.
Scalability	Architecture separates training and serving, allowing the dataset, feature set, or model to be extended without redesigning the web layer.

The scalability row deserves particular emphasis: because `train.py` and `app.py` never share runtime state, the dataset can be swapped, additional features engineered, or a fifth model added to the comparison in `train.py` without touching a single line of the Flask serving code, as long as the resulting scaler and model are re-saved under the same filenames in `models/`.

